

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

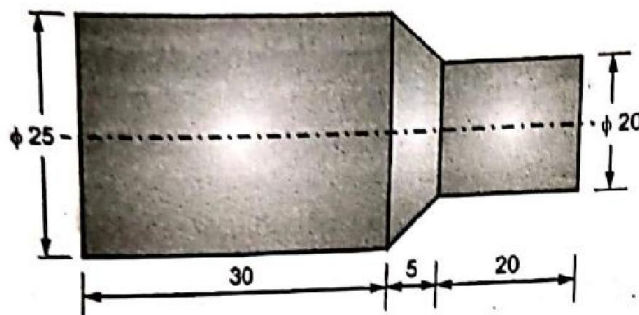
12

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|---|-----|---|
| A) Differentiate clearly between NC and CNC systems. Classify NC/CNC systems in detail. | CO1 | 6 |
| B) Write in short about recirculating ball screws used in CNC machines. | CO1 | 6 |
| C) What are the different feedback devices used in CNC machines? Explain any two in detail. | CO1 | 6 |

Q.2 Solve Any Two of the following.

12

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| A) Prepare a part program to machine the work piece as shown in the Figure. | CO2 | 6 |
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| B) Describe Automatic Tool Changers (ATC) used in CNC machines. | CO2 | 6 |
| C) Explain the work holding devices used in CNC machines. | CO2 | 6 |

Q. 3 Solve Any Two of the following.

12

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|---|-----|---|
| A) Discuss Electro-Chemical Machining process in detail. | CO3 | 6 |
| B) Compare Laser Beam Machining with Electron Beam Machining. | CO3 | 6 |
| C) Explain the process of Abrasive Jet Machining in detail. | CO3 | 6 |

Q.4 Solve Any Two of the following.		12
A) Explain the basic principles involved in thermal spraying with the help of schematic diagrams. What are the different types of thermal spraying methods?	CO4	6
B) Differentiate clearly between Chemical Vapour Deposition and Physical Vapour Deposition techniques. Explain the basic principles of both the processes using schematic diagrams.	CO4	6
C) Briefly describe the electroplating and electroless plating processes.	CO4	6
Q. 5 Solve Any Two of the following.		12
A) Explain the steps involved in rapid prototyping process.	CO5	6
B) Describe 3-D printing process in detail	CO5	6
C) Explain the LIGA micro-fabrication method with the help of schematic diagrams.	CO6	6

***** End *****